Yellampati Swathi
192324095

Section 1: Resume Information Extraction

Aim: Extract candidate details, identify technical skills, calculate eligibility, and generate a structured profile summary.

Program:

Python Program

```
import re

resume = """
Name: Rahul Kumar
Email: rahul@gmail.com
Mobile: 9876543210
Skills: Python, Java, SQL, Machine Learning, NLP
Experience: 3 years
"""

name = re.search(r"Name:\s*(.*)", resume)
email = re.findall(r"[\w.-]+@[\w.-]+\.\w+", resume)
mobile = re.findall(r"\b[6-9]\d{9}\b", resume)
experience = re.search(r"Experience:\s*(\d+)\s*years?", resume)

skills = ["Python", "Java", "SQL", "Machine Learning", "NLP"]
found_skills = []

for skill in skills:
    if re.search(re.escape(skill), resume, re.I):
        found_skills.append(skill)

years = int(experience.group(1)) if experience else 0
candidate_name = name.group(1).strip() if name else "Not found"

print("Candidate Name:", candidate_name)
print("Email:", email)
print("Mobile:", mobile)
print("Technical Skills:", found_skills)
print("Experience:", years, "years")
```

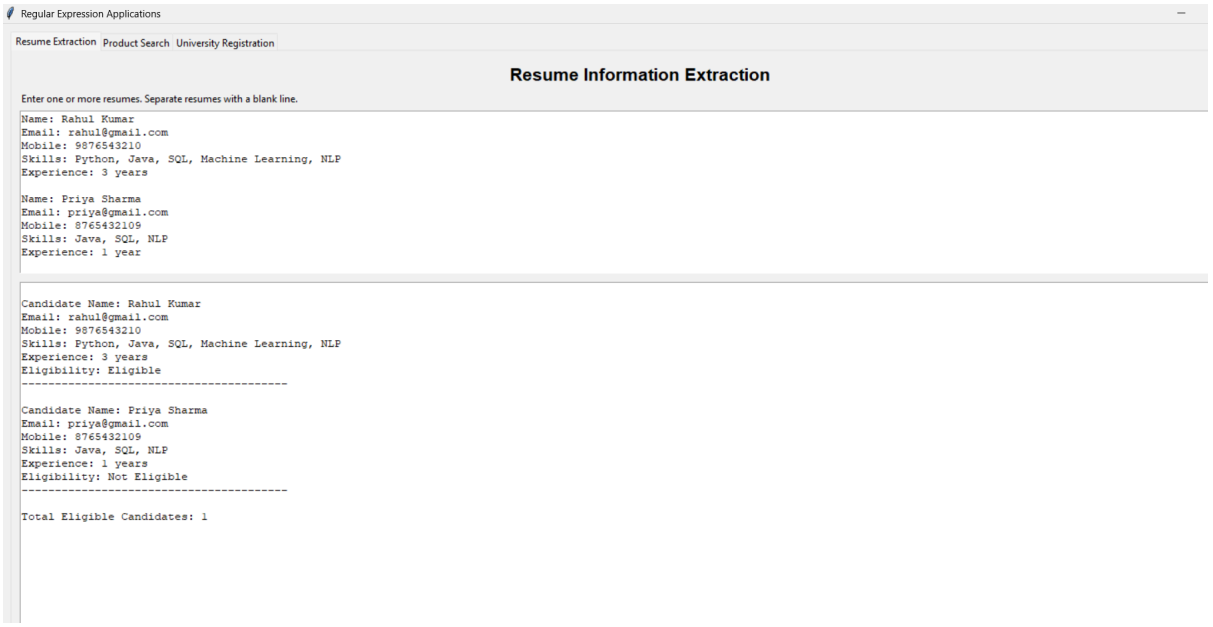
```
print("\nCandidate Profile Summary")
print("Name:", candidate_name)
print("Skills:", ", ".join(found_skills))
print("Experience:", years, "years")

if years >= 2 and "Python" in found_skills:
    print("Eligibility: Eligible")
else:
    print("Eligibility: Not Eligible")
```

Sample Output

```
Candidate Name: Rahul Kumar
Email: ['rahul@gmail.com']
Mobile: ['9876543210']
Technical Skills: ['Python', 'Java', 'SQL',
                   'Machine Learning', 'NLP']
Experience: 3 years

Candidate Profile Summary
Name: Rahul Kumar
Skills: Python, Java, SQL, Machine Learning, NLP
Experience: 3 years
Eligibility: Eligible
```



Result: The program extracts resume information and identifies candidates with at least 2 years of experience and Python skills.

Section 2: Product Search System

Aim: Search products using exact, prefix, suffix, partial, and case-insensitive matching and display the number of matching products.

Program:

Python Program

```
import re

products = [
    "Laptop",
    "Laptop Bag",
    "Smartphone",
    "Smart Watch",
    "Wireless Mouse",
    "Wireless Keyboard",
    "Laptop Stand"
]

keyword = input("Enter search keyword: ")

exact = []
prefix = []
suffix = []
partial = []

for product in products:
    if re.fullmatch(re.escape(keyword), product, re.I):
        exact.append(product)

    if re.match(re.escape(keyword), product, re.I):
        prefix.append(product)

    if re.search(re.escape(keyword) + r"$", product, re.I):
        suffix.append(product)

    if re.search(re.escape(keyword), product, re.I):
        partial.append(product)

print("\nExact Matches:", exact)
print("Prefix Matches:", prefix)
print("Suffix Matches:", suffix)
print("Partial Matches:", partial)
```

```
print("\nSearch Report")
print("Exact Matches:", len(exact))
print("Prefix Matches:", len(prefix))
print("Suffix Matches:", len(suffix))
print("Partial Matches:", len(partial))
```

Sample Output

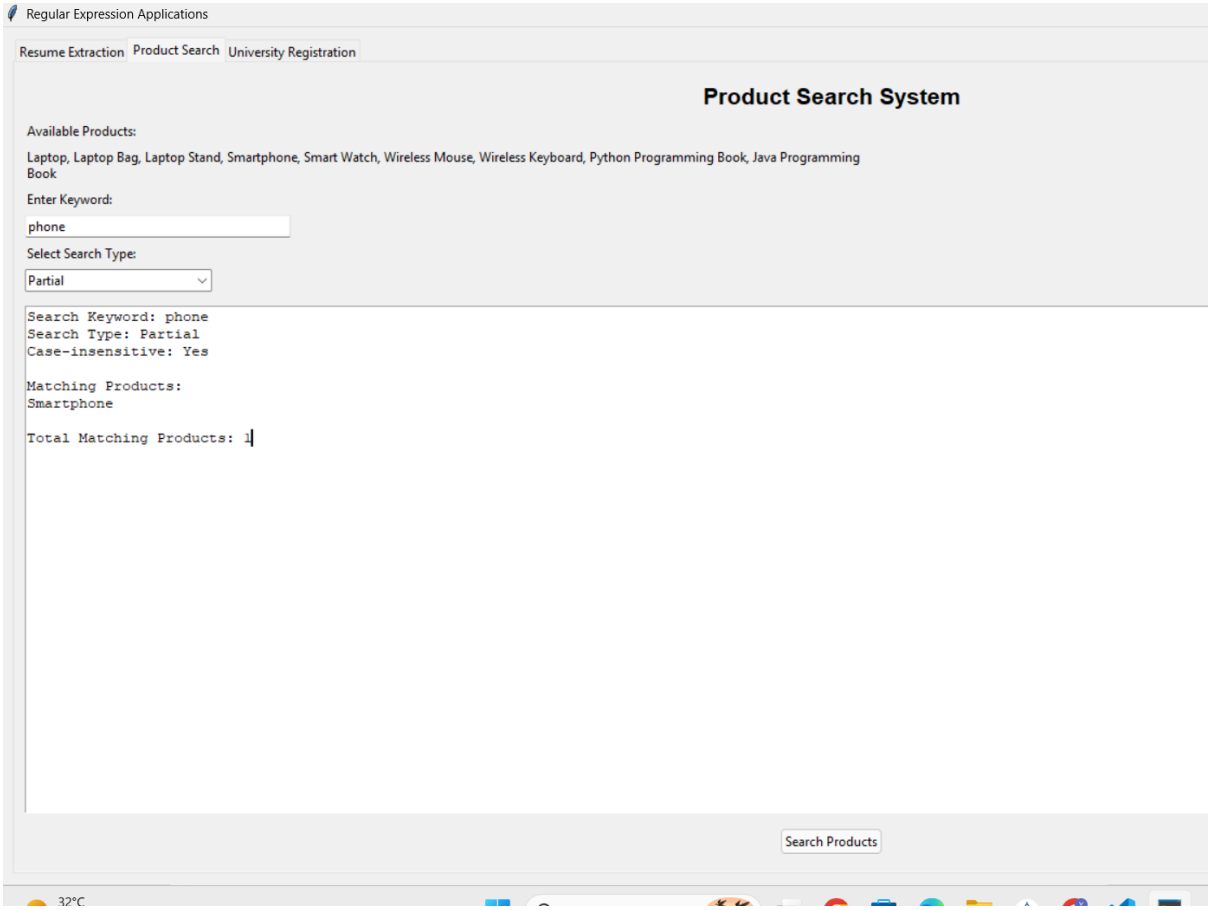
Input:

Enter search keyword: Laptop

Output:

Exact Matches: ['Laptop']
Prefix Matches: ['Laptop', 'Laptop Bag', 'Laptop Stand']
Suffix Matches: ['Laptop']
Partial Matches: ['Laptop', 'Laptop Bag', 'Laptop Stand']

Search Report
Exact Matches: 1
Prefix Matches: 3
Suffix Matches: 1
Partial Matches: 3



Result: The program searches products using different matching methods and generates a report of matching product counts.

Section 3: University Registration System

Aim: Validate the student register number, institutional email, course code, semester, and mobile number using Regular Expressions.

Program:

Python Program

```
import re

reg_no = input("Enter Register Number: ")
email = input("Enter Institutional Email: ")
course = input("Enter Course Code: ")
semester = input("Enter Semester: ")
mobile = input("Enter Mobile Number: ")

r1 = re.fullmatch(r"\d{2}[A-Z]{3}\d{4}", reg_no)
r2 = re.fullmatch(r"[\w.-]+@university\.edu", email)
r3 = re.fullmatch(r"[A-Z]{3}\d{3}", course)
r4 = re.fullmatch(r"[1-8]", semester)
r5 = re.fullmatch(r"[6-9]\d{9}", mobile)

print("\nValidation Report")

print("Register Number:",
      "Valid" if r1 else "Invalid")
print("Institutional Email:",
      "Valid" if r2 else "Invalid")
print("Course Code:",
      "Valid" if r3 else "Invalid")
print("Semester:",
      "Valid" if r4 else "Invalid")
print("Mobile Number:",
      "Valid" if r5 else "Invalid")

if r1 and r2 and r3 and r4 and r5:
    print("\nRegistration Status: Successful")
else:
    print("\nRegistration Status: Failed")
```

Sample Output

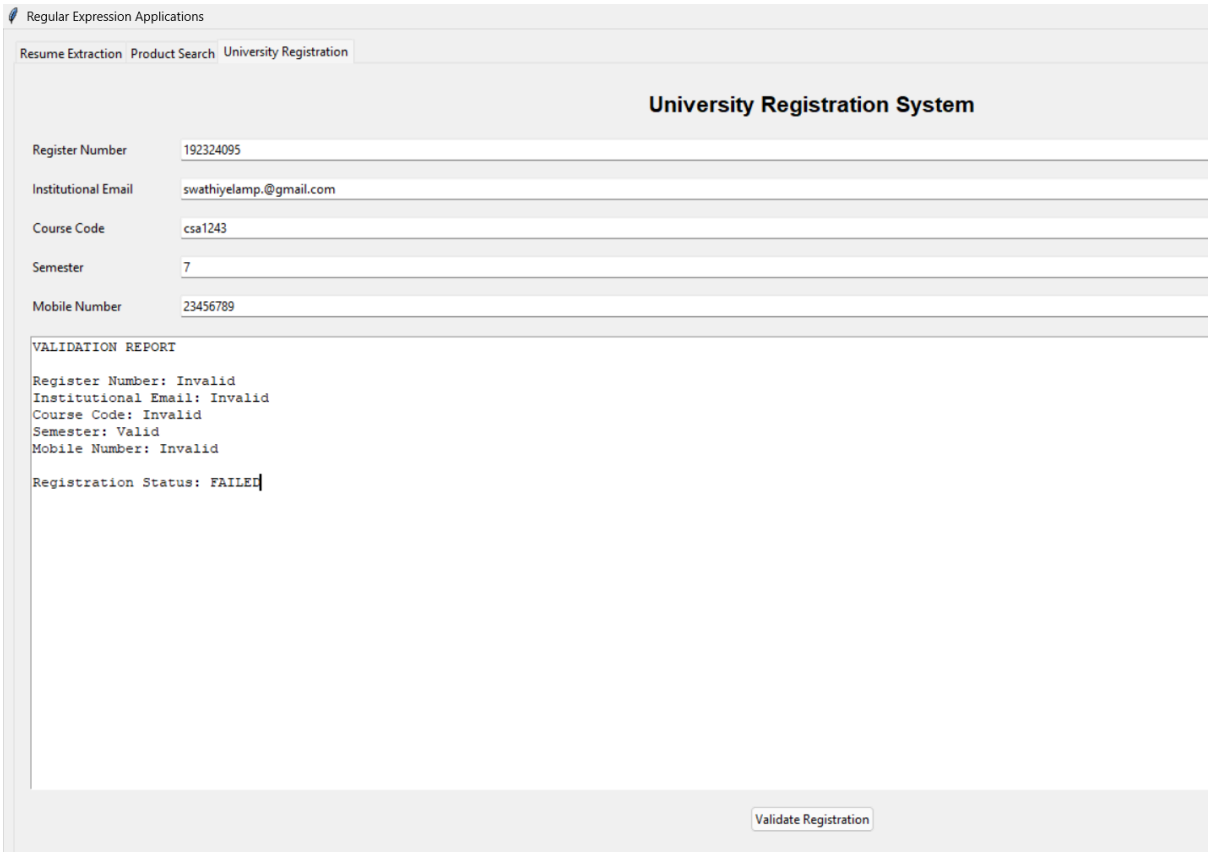
Input:

Enter Register Number: 22CSE1234
Enter Institutional Email: rahul@university.edu
Enter Course Code: CSE101
Enter Semester: 5
Enter Mobile Number: 9876543210

Output:

Validation Report
Register Number: Valid
Institutional Email: Valid
Course Code: Valid
Semester: Valid
Mobile Number: Valid

Registration Status: Successful



Result: The program validates all student registration fields and displays whether registration is successful.